6MMCS009W1 Usability Testing & Evaluation

Group Coursework

Group Number - 38

	First Name	Last Name	IIT ID	UOW ID
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BSc (Hons) in Computer Science

School of Computer Science & Engineering
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Experimental Operandum

Group information

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	Student Name	IIT ID	UOW ID
01	**Thalal Izzath**	20200165	w1868568
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Introduction and Objectives

This experiment will be conducted to assess the usability of the **Google Flights** platform. The evaluation will focus on its core features for searching, filtering, and selecting travel itineraries, measured in terms of key usability metrics: **Effectiveness**, **Efficiency**, **Error Tolerance**, and **Ease of Learning**.

As a group, our aim is to identify potential usability problems users encounter during the flight planning process and to understand how these issues impact their overall user experience.

As such, the research objectives are to:

1. Design and conduct a structured usability test to capture both **quantitative and qualitative data**.
2. Analyze users' interactions with the platform's key functionalities, such as its data-rich search results and advanced filtering tools.
3. Formulate **evidence-based recommendations** for improving the product's usability.

The present study aims to shed light on the accessibility and usability of Google Flights, particularly for **different user groups** with varying levels of travel-booking experience.

Research Questions and Hypotheses

In this section define the specific research questions and hypotheses the experiment aims to address.

Research Questions: Each group member should create a unique research question guiding the study (1 research question) focusing on different usability metrics for the same application/product/service.

Hypotheses: Each group member should formulate one hypothesis that corresponds to the research question they created, detailing expected relationships between variables. Each hypothesis should align with one of the following statistical tests:

- *Independent-sample t-test*
- *Paired-sample t-test*
- *ANOVA*
- *Pearson correlation*

As a result, the team should collectively have four research questions, each with a hypothesis that corresponds to one of the statistical tests listed above.

Each group member is responsible to complete the sections below to describe their contribution to formulating the research questions and hypothesis of this usability testing session.

]

2.1 Independent-sample-test

[Provide the full name of the team member responsible for preparing the study to collect data that will be studied with Independent-sample-t-test.
]

Student Name: Anika

[Use the template below to provide your individual contribution to setting up this usability testing session. Replace the instructions with your response. Remove these instructions from the submitted report.]

Research question	*Ensure the question clearly defines what you aim to evaluate, such as a specific aspect of usability*	
Usability metric studied	*Formulate the hypothesis to directly address the research question with measurable outcomes*	
Hypothesis (statement)	*Null Hypothesis (H_0):*	
	Alternative Hypothesis (H_1):	
Mathematical Expression	*Null Hypothesis (H_0):*	
	Alternative Hypothesis (H_1):	
Variables	*Independent*	*the variable in an experiment that is manipulated by the researcher*
	Dependent	*what is measured*
	Control	*Task given*
	Measurement Scale	*e.g., Likert scales for satisfaction, time in seconds for completion*
	Data type	*What data type is it?*

Experimental design	*What experimental design you used: between-subjects, within-subjects, or mixed design/split-plot and why?*
Tasks and Scenarios	*What task the participants should complete to allow the collection of data to evaluates the hypothesis?*
Procedure	*What procedure you used to collect the data: in person, remotely, or in a naturalistic setting, and if will be moderated or un-moderated with justification about your choice.*
Data collection method	*Identify the tools you will use for accurate data collection and justify why those are best fit to the collecting the data for this hypothesis*
Test material	*What material you used to collect data for this hypothesis (e.g. which question of the user questionnaire), instructions to participants for the task to be completed, how this is measured (e.g. if you record task time define task start and end)*

2.2 Paired-sample-test

[Provide the full name of the team member responsible for preparing the study to collect data that will be studied with Independent-sample-t-test.
]

Student Name: Thalal

[Use the template below to provide your individual contribution to setting up this usability testing session. Replace the instructions with your response. Remove these instructions from the submitted report.]

Research question	*Ensure the question clearly defines what you aim to evaluate, such as a specific aspect of usability*	
Usability metric studied	*Formulate the hypothesis to directly address the research question with measurable outcomes*	
Hypothesis (statement)	*Null Hypothesis (H_0):*	
	Alternative Hypothesis (H_1):	
Mathematical Expression	*Null Hypothesis (H_0):*	
	Alternative Hypothesis (H_1):	
Variables	*Independent*	*the variable in an experiment that is manipulated by the researcher*
	Dependent	*what is measured*
	Control	*Task given*
	Measurement Scale	*e.g., Likert scales for satisfaction, time in seconds for completion*
	Data type	*What data type is it?*
Experimental design	*What experimental design you used: between-subjects, within-subjects, or mixed design/split-plot and why?*	
Tasks and Scenarios	*What task the participants should complete to allow the collection of data to evaluates the hypothesis?*	
Procedure	*What procedure you used to collect the data: in person, remotely, or in a naturalistic setting, and if will be moderated or un-moderated with justification about your choice.*	
Data collection method	*Identify the tools you will use for accurate data collection and justify why those are best fit to the collecting the data for this hypothesis*	

Test material	*What material you used to collect data for this hypothesis (e.g. which question of the user questionnaire), instructions to participants for the task to be completed, how this is measured (e.g. if you record task time define task start and end)*

2.3 ANOVA

[Provide the full name of the team member responsible for preparing the study to collect data that will be studied with Independent-sample-t-test.
]

Student Name: Saad

[Use the template below to provide your individual contribution to setting up this usability testing session. Replace the instructions with your response. Remove these instructions from the submitted report.]

Research question	*Ensure the question clearly defines what you aim to evaluate, such as a specific aspect of usability*	
Usability metric studied	*Formulate the hypothesis to directly address the research question with measurable outcomes*	
Hypothesis (statement)	*Null Hypothesis (H_0):*	
	Alternative Hypothesis (H_1):	
Mathematical Expression	*Null Hypothesis (H_0):*	
	Alternative Hypothesis (H_1):	
Variables	*Independent*	*the variable in an experiment that is manipulated by the researcher*
	Dependent	*what is measured*
	Control	*Task given*
	Measurement Scale	*e.g., Likert scales for satisfaction, time in seconds for completion*
	Data type	*What data type is it?*
Experimental design	*What experimental design you used: between-subjects, within-subjects, or mixed design/split-plot and why?*	
Tasks and Scenarios	*What task the participants should complete to allow the collection of data to evaluates the hypothesis?*	
Procedure	*What procedure you used to collect the data: in person, remotely, or in a naturalistic setting, and if will be moderated or un-moderated with justification about your choice.*	
Data collection method	*Identify the tools you will use for accurate data collection and justify why those are best fit to the collecting the data for this hypothesis*	
Test material	*What material you used to collect data for this hypothesis (e.g. which question of the user questionnaire), instructions to participants for the task to be completed, how this is measured (e.g. if you record task time define task start and end)*	

2.4 Pearson Correlation

[Provide the full name of the team member responsible for preparing the study to collect data that will be studied with Independent-sample-t-test.

]

Student Name: Linushan

[Use the template below to provide your individual contribution to setting up this usability testing session. Replace the instructions with your response. Remove these instructions from the submitted report.]

Research question	*Ensure the question clearly defines what you aim to evaluate, such as a specific aspect of usability*	
Usability metric studied	*Formulate the hypothesis to directly address the research question with measurable outcomes*	
Hypothesis (statement)	*Null Hypothesis (H_0):*	
	Alternative Hypothesis (H_1):	
Mathematical Expression	*Null Hypothesis (H_0):*	
	Alternative Hypothesis (H_1):	
Variables	*Independent*	*the variable in an experiment that is manipulated by the researcher*
	Dependent	*what is measured*
	Control	*Task given*
	Measurement Scale	*e.g., Likert scales for satisfaction, time in seconds for completion*
	Data type	*What data type is it?*
Experimental design	*What experimental design you used: between-subjects, within-subjects, or mixed design/split-plot and why?*	
Tasks and Scenarios	*What task the participants should complete to allow the collection of data to evaluates the hypothesis?*	
Procedure	*What procedure you used to collect the data: in person, remotely, or in a naturalistic setting, and if will be moderated or un-moderated with justification about your choice.*	
Data collection method	*Identify the tools you will use for accurate data collection and justify why those are best fit to the collecting the data for this hypothesis*	
Test material	*What material you used to collect data for this hypothesis (e.g. which question of the user questionnaire), instructions to participants for the task to be completed, how this is measured (e.g. if you record task time define task start and end)*	

3. Study planning and organisation - ANIKA

[Remove these instructions from the submitted report.

In this section explain how you organised the study to investigate the formulated research questions and hypotheses as a group.

Although this section is written as a group, each member is responsible for contributing data collection details related to the research questions they individually developed.

You are expected to discuss the following:

3.1 Experimental Design - ANIKA

3.1.1 **Experimental Design for Research Question 1 (Anika)**

3.1.2 **Experimental Design for Research Question 2 (Thalal)**

3.1.3 **Experimental Design for Research Question 3 (Saad)**

3.1.4 **Experimental Design for Research Question 1 (Linushan)**

*Describe the chosen **experimental design** (between-subjects, within-subjects, or mixed design/split-plot) to address each formulated hypothesis. Since different hypotheses may require different designs, explain how you structured your study to effectively collect data for each hypothesis.*

3.2 Procedure and Planning - ANIKA

3.2.1. **Independent-Sample Test – In-Person (Moderated Unmoderated) Task (Anika)**
3.2.2. **Paired-Sample Test – In-Person (Moderated Unmoderated) Task (Thalal)**
3.2.3. **ANOVA – In-Person (Moderated Unmoderated) Task (Saad)**
3.2.4. **Pearson Correlation – In-Person (Moderated Unmoderated) Task (Linushan)**
3.2.5. **Participants Information Sheet (Anika)**

Outline the step-by-step procedure followed during each session, including task instructions, timing, and any standardization for consistency.

- ***Test Duration**: Describe the duration of each test session and the overall testing period. For example, each session might have lasted 30 minutes, including time for welcoming participants, explaining the study process, conducting the test, and collecting self-reported data through a post-test questionnaire. The entire test period may have spanned one week to collect data from 20 participants.*
- *What **tasks the participants** had to complete and why? How those tasks allowed the collection of data to evaluate individual's test hypothesis?*
  - ***Task/s for Independent-Sample Test Hypothesis**: Describe the tasks users completed to collect data for evaluating the independent-sample test hypothesis.*
  - ***Task/s for Paired-Sample Test Hypothesis**: Describe the tasks users completed to collect data for evaluating the paired-sample test hypothesis.*
  - ***Task/s for ANOVA Hypothesis**: Describe the tasks users completed to collect data for evaluating the hypothesis analyzed with ANOVA.*
  - ***Task/s for Correlational Hypothesis**: Describe the tasks users completed to collect data for evaluating the correlational hypothesis.*
- ***Provide participant's information sheet.** If it was unmoderated what was the process you followed to ensure that the participants knew what to do and that data was recorded with precision? From this section provide an internal reference to **Appendix I.** Participant Information Sheet.*

3.3 Roles & Responsibilities - ANIKA

3.3.1. **Independent-Sample Test – Roles & Responsibilities (Anika)**
3.3.2. **Paired-Sample Test – Roles & Responsibilities (Thalal)**
3.3.3. **ANOVA – Roles & Responsibilities (Saad)**
3.3.4. **Pearson Correlation –Roles & Responsibilities (Linushan)**

Outline the roles of those involved in the usability test (e.g., moderators, facilitators) and explain how each group member contributed to meeting the study's requirements and facilitating the sessions effectively.

- *What was the instructions that the moderator had to follow? From this section provide an internal reference to* ***Appendix II.*** Moderator Information Sheet
- ***Planning for the Usability Session****: Describe the preparations required to conduct the usability session. For example, did you need to book a lab or reserve a room at the library? Did you arrange for laptops or set up devices with the necessary software? Explain whether you recruited participants to complete the test over multiple days and describe any other logistical steps you took to ensure the session ran smoothly.*

3.4 Participant Profile and Recruitment

3.4.1. **Independent-Sample Test – Participant Profile and Recruitment (Anika)**
3.4.2. **Paired-Sample Test –Participant Profile and Recruitment (Thalal)**
3.4.3. **ANOVA – Participant Profile and Recruitment (Saad)**
3.4.4. **Pearson Correlation –Participant Profile and Recruitment (Linushan)**

Participant Profile and Recruitment*: Provide a detailed description of the participants' profiles (e.g., age, experience level) and explain why this specific profile is relevant to the study's objectives.*

Describe the ***recruitment methods*** *used and include* ***evidence of any recruitment materials*** *(e.g., flyers, emails, or online posts). From this section provide an internal reference to* Appendix III. Participant Recruitment Evidence.

3.5 Data Collection Methods

3.4.1. **Independent-Sample Test – Data Collection Methods (Anika)**
3.4.2. **Paired-Sample Test –Data Collection Methods (Thalal)**
3.4.3. **ANOVA – Data Collection Methods (Saad)**
3.4.4. **Pearson Correlation –Data Collection Methods (Linushan)**

- *Explain how data was collected, including quantitative and qualitative data, and any tools used for recording responses or observations.*
  - *Where the study took place:* ***in person, remotely****?*

- *Was the study **moderated, or un-moderated**? If it was moderated what was the role of the instructor?*

3.6 Research Instruments (materials) and Equipment

*What **tools/test material you used to collect the data**? Specify all materials and equipment used, such as the prototype you used, software, hardware, questionnaires, or data collection tools for accurate data collection and session recording and justify why those are best fit for the test session you are designing. From this section provide an internal reference to **Appendix IV.*** Data Collection Material *(if you used a questionnaire to collect data).*

3.8 Pilot Testing

If applicable, describe any pilot testing conducted to refine the experimental setup and ensure procedures work as intended.

]

Report the study raw data

[Remove these instructions from the submitted report.

In this section present the results in a table and explain what they represent and how they are related to the research questions you have formed, and you will use to conduct and the variables of the hypothesis you study

- *independent-sample-test*
- *paired-sample t-test*
- *ANOVA*
- *Pearson correlation r-test*

*From this section provide an internal reference to **Appendix VI**.* **Raw data** *and evidence of test recordings* Appendix V. Test Recordings.

]

Ethics form

[Remove these instructions from the submitted report.

As a group discuss ethical practices followed, such as obtaining informed consent, ensuring participant anonymity, and data protection measures.

As a group you complete and submit
Part-A-Research-Ethics-Consideration-Application-form2020_21.

https://www.westminster.ac.uk/research/research-governance/research-and-knowledge-exchange-ethics

*From this section provide a reference to the Ethics form in **Appendix VII**.* Ethics form
]

Conclusions

[Remove these instructions from the submitted report.

Be reflective on the testing process you have followed. What you could have done better. Did you do a pilot and if so how this helped you to improve the testing process?

Identify any limitations in the experimental setup and steps taken to control for potential biases or external influences.

]

Appendix

Appendix I. Participant Information Sheet

[Remove these instructions from the submitted report.

Provide here the participants' task list.

]

Appendix II. Moderator Information Sheet

[Remove these instructions from the submitted report.

Provide here the instructors Information Sheet.

]

Appendix III. Participant Recruitment Evidence

[Remove these instructions from the submitted report.

Provide here evidence of any recruitment materials (e.g., flyers, emails, or online posts).

]

Appendix IV. Data Collection Material

[Remove these instructions from the submitted report.

Provide here a link to an online Questionnaire you use or other material you have created to collect data.

]

Appendix V. Test Recordings

[Remove these instructions from the submitted report.

Provide here evidence for the data collection process, like recordings of test sessions. This could be a link to a youtube video that shows the test recording of a test session. Be aware of GDPR.

]

Appendix VI. Raw data

[Remove these instructions from the submitted report.

Provide here a link to raw data of collected the results (e.g. completed Questionnaires, moderators notes, recordings).

]

Appendix VII. Ethics form

[Remove these instructions from the submitted report.

Provide here instructions of accessing the Ethics form.

]